

JAZIM SYED

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SUMMARY OF QUALIFICATIONS

- Practical experience in data analysis using Excel and Python, and a keen interest in clinical research and regulatory compliance
- Proven collaboration and communication skills, demonstrated through cross-functional teamwork, precise data management, and clear documentation
- Familiar with medical technology and ultrasound imaging concepts, with a strong commitment to supporting quality assurance and patient-focused healthcare solutions
- Adaptable, detail-oriented, and highly motivated to contribute in a hybrid clinical research environment, working alongside data science and machine learning teams

Software: C, C++, Python, Matlab, Solidworks, Quartus II, NI Multisim, Proteus 8.11, MPLAB X IDE, MS Office

EDUCATION

Biomedical Engineering (B. Eng), Toronto Metropolitan University (Formerly Ryerson)

EXP. JUN 2026

Relevant Courses: Microprocessors, Material Science, Electrical Systems, Control Systems, Biology, Biomechanics

PROFESSIONAL EXPERIENCE

Personal Support Worker, *Extendicare Long Term Care Home*

AUG 2023 - PRESENT

- Monitored and logged resident behavior every 30 minutes, enabling data-driven decisions in collaboration with nursing staff to adjust medications and provide additional support, improving overall resident well-being
- Led COVID-19 protocol enforcement for residents, staff, and 75+ visitors, implementing screening measures and symptom checks that helped maintain health standards and prevent onsite outbreaks

PERSONAL PROJECTS

Developed an Arduino Based Breathalyzer with MQ-3 Sensor

2025

- Designed and built a low-cost breathalyzer prototype using an Arduino Uno Microcontroller and MQ-3 alcohol sensor, achieving a calibration value of 0.96 for alcohol vapor concentration detection
- Incorporated a real-time alcohol monitoring system integrating user-specific factors (body weight, sex, time since last drink) to enhance BAC estimation precision
- Optimized system costs to approximately \$30 CAD total, delivering a 70% cheaper alternative to commercial breathalyzers without significant loss in reliability

Machine Learning Model for Kidney Cancer Detection

2024

- Designed and trained a machine learning model that achieved 92% accuracy in detecting kidney cancer from a dataset of over 10,000 CT and PET scans, showcasing strong high-speed data processing and analysis skills
- Applied generative AI techniques and advanced data augmentation to increase dataset diversity, enhancing the model's ability to accurately differentiate between cancerous and non-cancerous scans.

Temperature Sensor Interfacing and Peripheral Control

2024

- Interfaced a TMP36 analog temperature sensor with a PIC24 microcontroller using Proteus 8.11, achieving real-time temperature monitoring with an ADC resolution of 0.488°C per step
- Configured 10-bit ADC, UART communication (19200 baud rate), and Timer 2 to enable accurate sensor data acquisition and real-time display on the Virtual Terminal.
- Programmed dynamic control of a DC fan based on temperature thresholds, with fan direction toggling at 30°C to stimulate biomedical thermal regulation systems

Post-Stroke Rehabilitation Exoskeleton

2023

- Developed a 3D model of a rehabilitation exoskeleton for post-stroke patients using Autodesk Inventor. Theorized the required fabrics and materials to build the exoskeleton. Incorporated the use of a PCB board for the circuit design.
- Overcame challenges of integrating a power system with no external charging through the use of a renewable energy source

EXTRACURRICULAR EXPERIENCE

VP Finance & Sponsorship Associate, Biomedical Engineering Course Union (BECU) - TMU

SEPT 2022- PRESENT

- Spearheaded the planning and execution of BMEC 2023, attracting 40–50 attendees and securing keynote speakers from Sanofi, SOTI, and graduate research programs, enhancing industry exposure and networking opportunities for students
- Managed a \$2,000 annual budget for BECU by maintaining detailed tracking of all income and expenditures through organized spreadsheets, ensuring financial transparency and fiscal responsibility throughout the academic year for a 200+ student organization